MATH 130 · UNIT 3A

Angles and Right Triangle Trigonometry

Concepts, worked examples, student practice, and synthesis

75 minutes · Offline capable

Today’s Learning Roadmap

1. sketch angles in standard position
2. find coterminal and reference angles in degrees and radians
3. use radians to compute arc length
4. use right-triangle trigonometric ratios
5. solve right triangles and word problems, including two-triangle problems

Predict: sketch angles in standard position

PREDICT FIRST

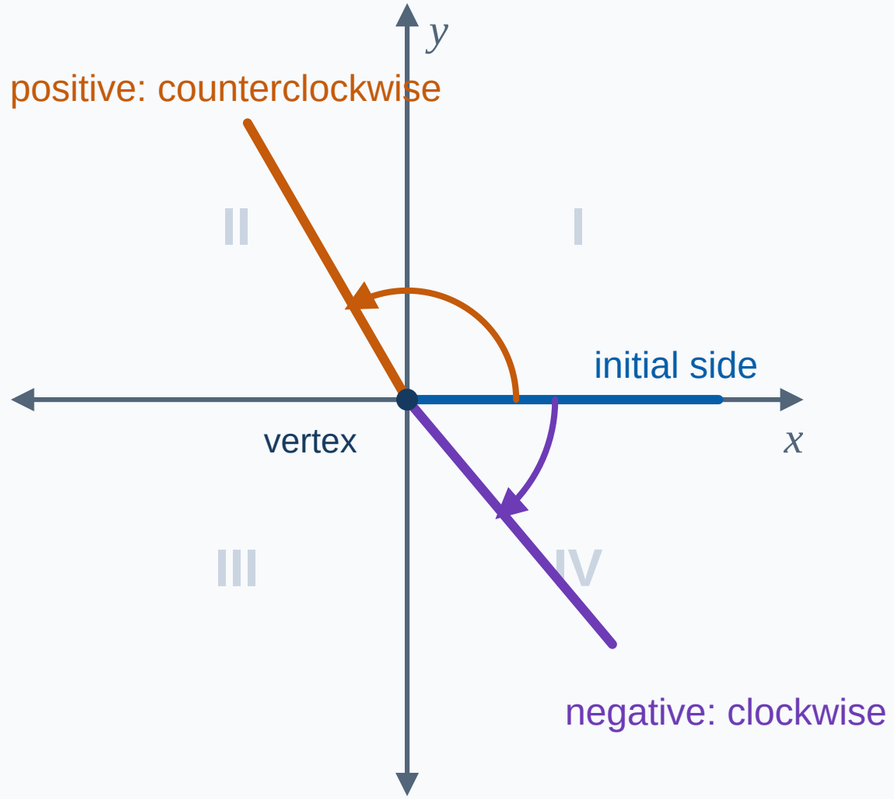
Where does a negative angle rotate, and what never changes about standard position?

Commit to a response before discussion.

sketch angles in standard position

CONCEPT

- The initial side lies on the positive x-axis.
- Positive rotation is counterclockwise; negative rotation is clockwise.



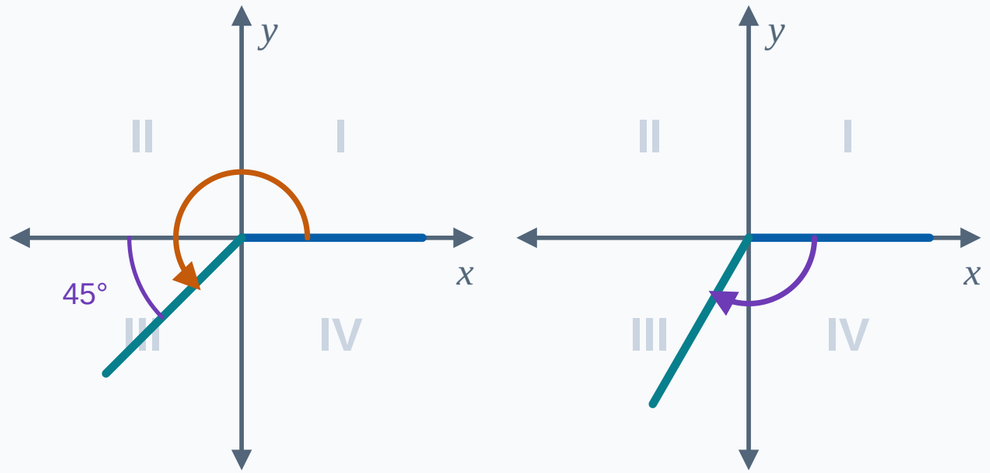
Start on the positive x-axis; the sign sets the direction.

Source: 2

Model the Skill: sketch angles in standard position

Sketch 225° and -120° and identify their quadrants.

1. $225^\circ = 180^\circ + 45^\circ$, so it terminates in Quadrant III.
2. $-120^\circ + 360^\circ = 240^\circ$, also in Quadrant III.
3. Both begin on the positive x-axis; rotation direction distinguishes the sketches.



225° = 180° + 45°

−120° (same side as 240°)

Both angles end in Quadrant III.

You Try: sketch angles in standard position

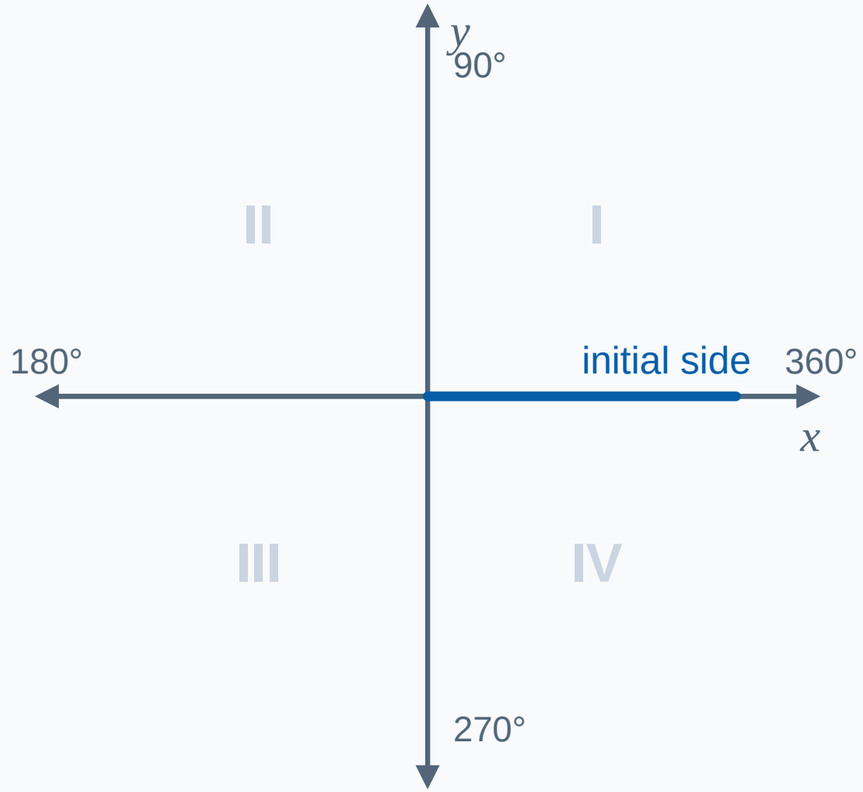
STUDENT PRACTICE

Sketch 310° and label initial and terminal sides.

Reveal solution

FEEDBACK AND SOLUTION

The initial side is the positive x-axis; 310° terminates in Quadrant IV, 50° below the positive x-axis.



Sketch 310° here.

Start on the initial side and rotate counterclockwise.

Check and Diagnose: sketch angles in standard position

ERROR ANALYSIS

- Do not measure from the y-axis.
- Quadrant boundaries are not inside a quadrant.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: sketch angles in standard position

PACING CHECKPOINT

Describe standard position without drawing it.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Source: Authored

Predict: find coterminal and reference angles in degrees and radians

PREDICT FIRST

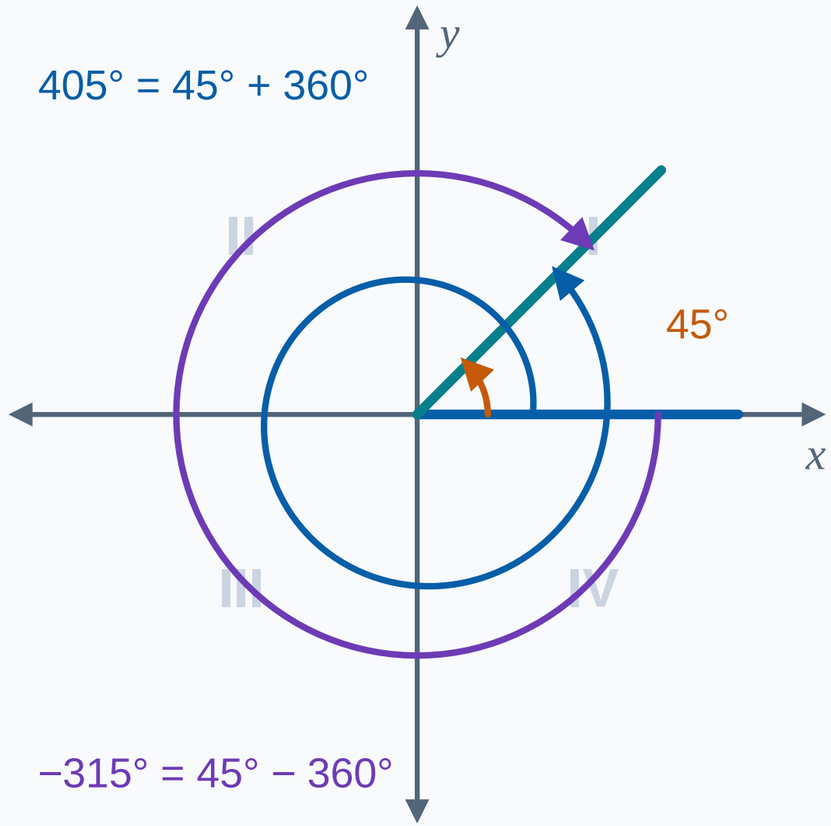
How can infinitely many angles share one terminal side?

Commit to a response before discussion.

find coterminal and reference angles in degrees and radians

FORMULA FOCUS

- Coterminal angles differ by $360^\circ k$ or $2\pi k$.
- Reduce large magnitudes before locating the terminal side.

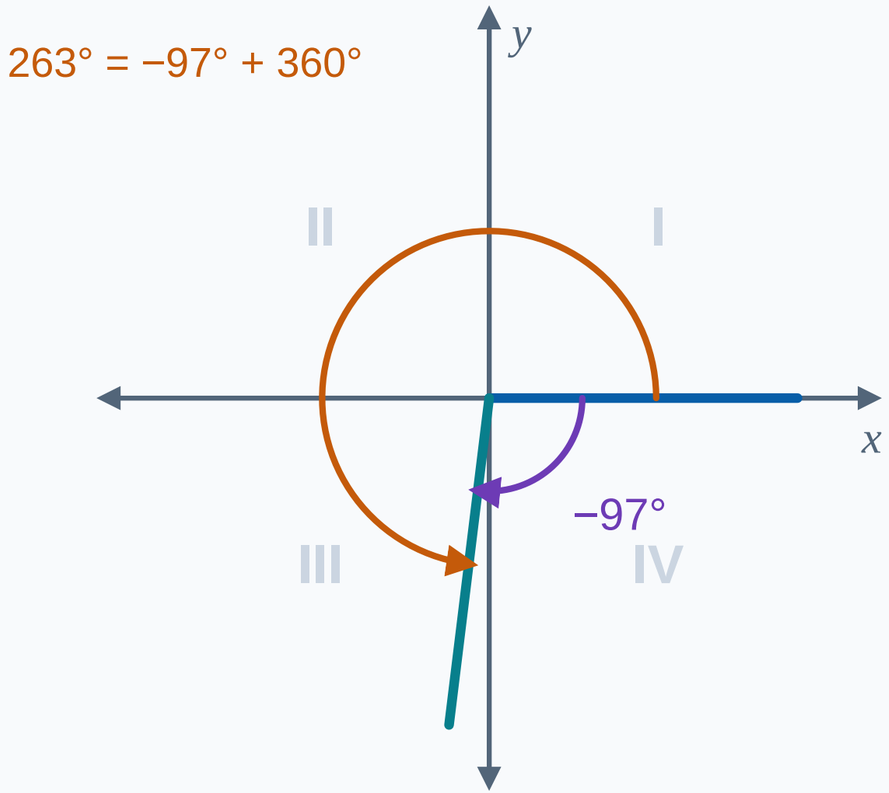


Full turns change the angle, not the terminal side.

Model the Skill: find coterminal and reference angles in degrees and radians

Find positive and negative coterminal angles for -97° and $9\pi/7$.

1. $-97^\circ + 360^\circ = 263^\circ$ and $-97^\circ - 360^\circ = -457^\circ$.
2. $9\pi/7 + 2\pi = 23\pi/7$ and $9\pi/7 - 2\pi = -5\pi/7$.
3. Coterminal angles differ by full rotations.



One terminal side, two directions of rotation.

You Try: find coterminal and reference angles in degrees and radians

STUDENT PRACTICE

Find a coterminal angle between 0° and 360° for 765°.

Reveal solution

FEEDBACK AND SOLUTION

$765 - 720 = 45^\circ$.

Check and Diagnose: find coterminal and reference angles in degrees and radians

ERROR ANALYSIS

- Add or subtract full rotations only.
- Keep degree and radian units separate.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: find coterminal and reference angles in degrees and radians

PACING CHECKPOINT

Give the general coterminal-angle formulas.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Source: Authored

Predict: use radians to compute arc length

PREDICT FIRST

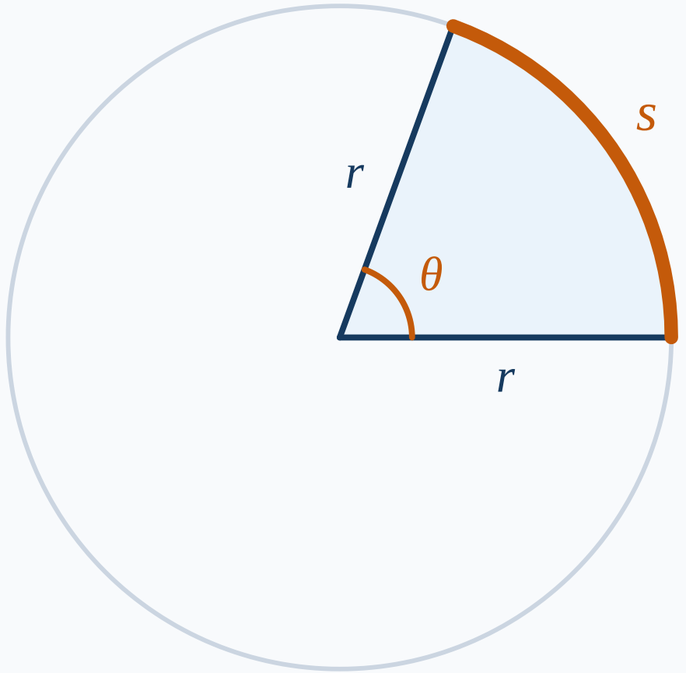
What does one radian measure geometrically?

Commit to a response before discussion.

use radians to compute arc length

FORMULA FOCUS

- Arc length is $s = r\theta$ when θ is in radians.
- Radians connect central angle directly to distance.



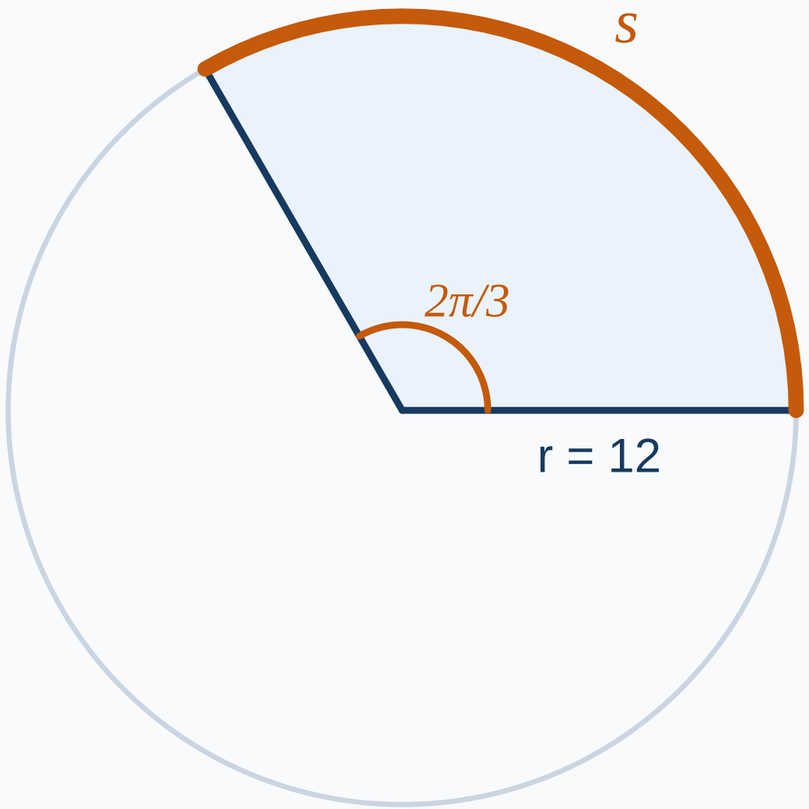
s = rθ (θ in radians)

The arc s sits opposite the central angle θ.

Model the Skill: use radians to compute arc length

For $r = 12$ and $\theta = 2\pi/3$, compute arc length.

1. The angle is already in radians.
2. $s = r\theta = 12(2\pi/3) = 8\pi$ units.
3. 8π is about 25.133 units, less than the full circumference 24π.



Two thirds of π is one third of a full turn.

Source: Authored

You Try: use radians to compute arc length

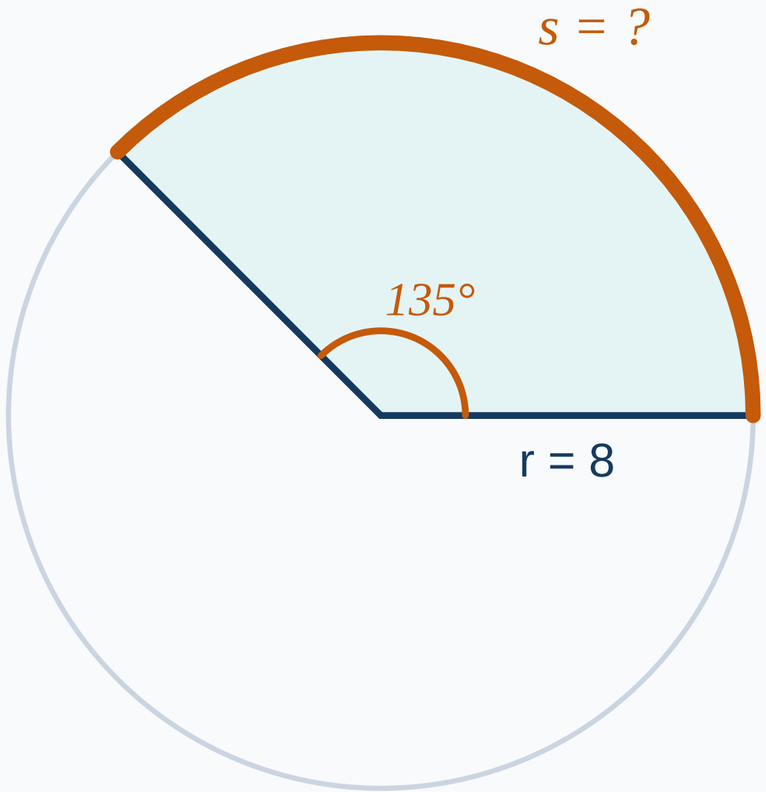
STUDENT PRACTICE

Find s for $r = 8$ and $\theta = 135^\circ$.

Reveal solution

FEEDBACK AND SOLUTION

$135^\circ = 3\pi/4$, so $s = 6\pi$.



Convert 135° to radians before using s = rθ.

Check and Diagnose: use radians to compute arc length

ERROR ANALYSIS

- Convert degrees before using $s = r\theta$.
- Arc length has linear units.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Source: Authored

Make It Stick: use radians to compute arc length

PACING CHECKPOINT

Explain why radians make $s = r\theta$ dimensionally natural.

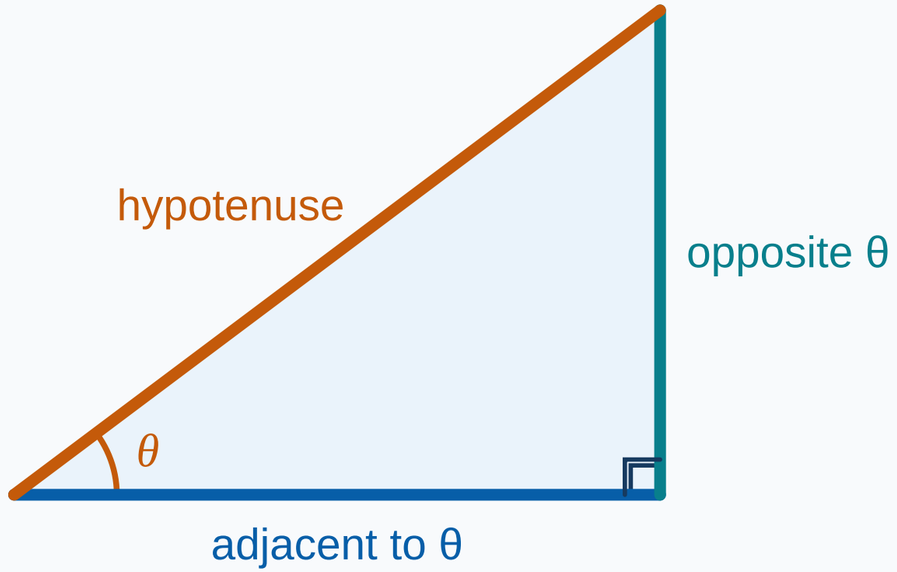
Exit response
Answer in complete mathematical sentences and identify one remaining question.

Predict: use right-triangle trigonometric ratios

PREDICT FIRST

Which two sides define sine, cosine, and tangent relative to an angle?

Commit to a response before discussion.

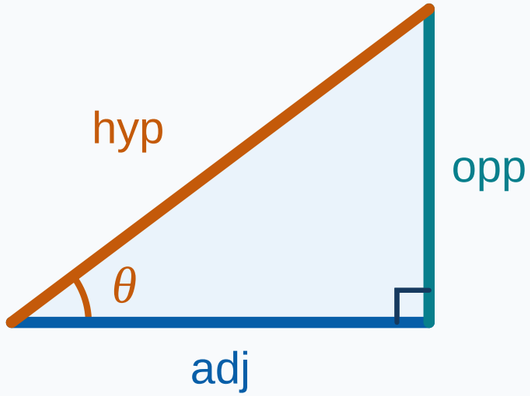


Each side is named from the angle θ.

use right-triangle trigonometric ratios

CONCEPT

- SOH-CAH-TOA selects the ratio.
- Reciprocal functions invert sine, cosine, and tangent.



SOH · CAH · TOA

reciprocals: flip

$\sin\theta = \frac{\text{opp}}{\text{hyp}}$ $\quad \csc\theta = \frac{\text{hyp}}{\text{opp}}$

$\cos\theta = \frac{\text{adj}}{\text{hyp}}$ $\quad \sec\theta = \frac{\text{hyp}}{\text{adj}}$

$\tan\theta = \frac{\text{opp}}{\text{adj}}$ $\quad \cot\theta = \frac{\text{adj}}{\text{opp}}$

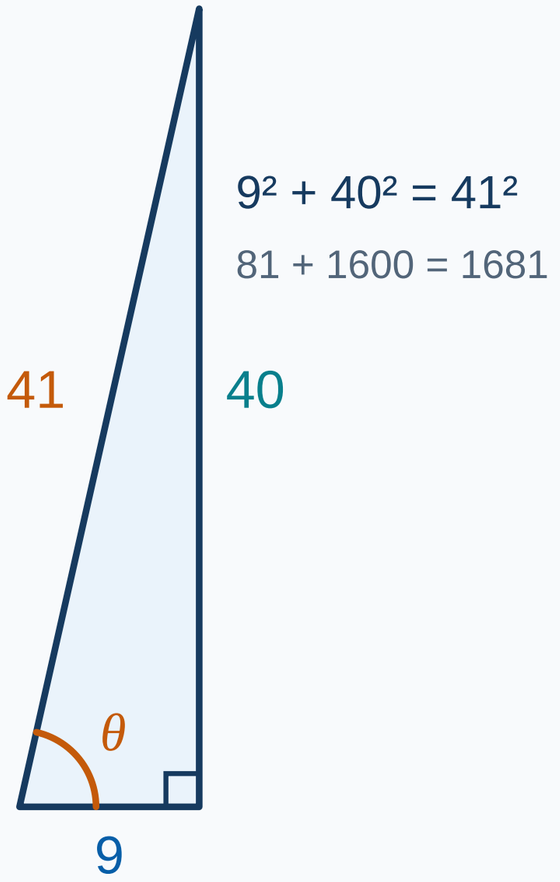
Colors match each side to its place in the ratios.

Source: 11, 12

Model the Skill: use right-triangle trigonometric ratios

A right triangle has opposite leg 40, adjacent leg 9, and hypotenuse 41. Find all six ratios.

1. $\sin\theta = 40/41$, $\cos\theta = 9/41$, and $\tan\theta = 40/9$.
2. $\csc\theta = 41/40$, $\sec\theta = 41/9$, and $\cot\theta = 9/40$.
3. $9^2 + 40^2 = 41^2$ verifies the side lengths.



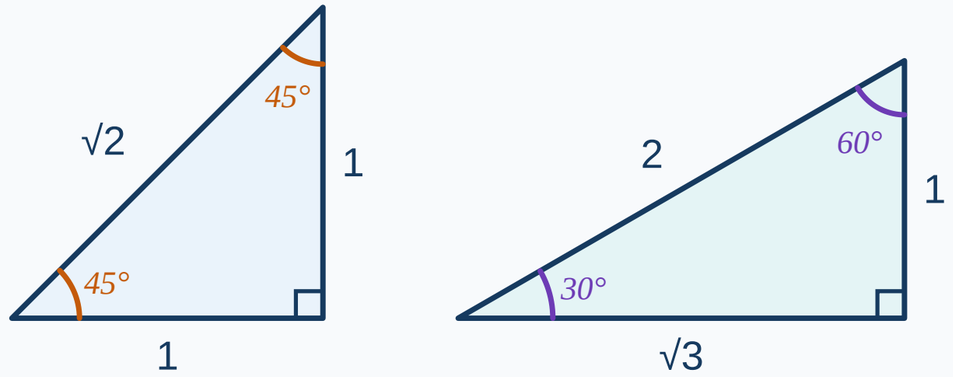
Drawn to scale: 9, 40, 41.

Source: 13, 14

Exact values from the special triangles

FORMULA FOCUS

- $\sin 30^\circ = \frac{1}{2}, \cos 30^\circ = \frac{\sqrt{3}}{2}, \tan 30^\circ = \frac{\sqrt{3}}{3}$
- $\sin 45^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}, \tan 45^\circ = 1$
- $\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \tan 60^\circ = \sqrt{3}$
- Keep exact values unless a decimal is asked for, and round only the final answer.



45°–45°–90°

30°–60°–90°

Exact side ratios for the two special triangles.

Source: Authored

You Try: use right-triangle trigonometric ratios

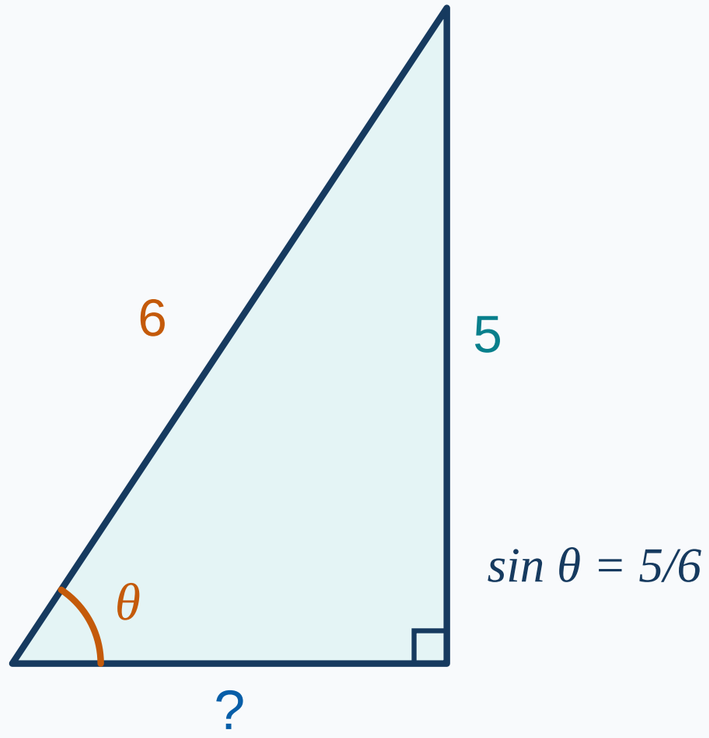
STUDENT PRACTICE

Given $\sin\theta = \frac{5}{6}$ for an acute angle θ, sketch the triangle, find the third side, and give the other five ratios as exact values.

Reveal solution

FEEDBACK AND SOLUTION

- Adjacent side: $\sqrt{6^2 - 5^2} = \sqrt{11}$.
- $\cos\theta = \frac{\sqrt{11}}{6}$, $\tan\theta = \frac{5}{\sqrt{11}} = \frac{5\sqrt{11}}{11}$, $\csc\theta = \frac{6}{5}$, $\sec\theta = \frac{6}{\sqrt{11}} = \frac{6\sqrt{11}}{11}$, $\cot\theta = \frac{\sqrt{11}}{5}$.



Givens only: find the third side first.

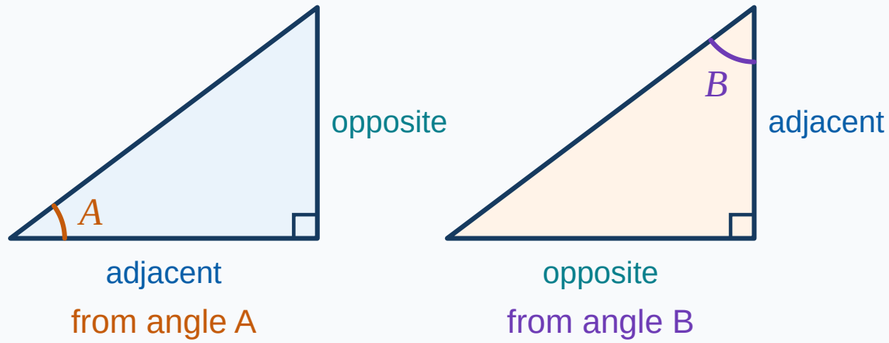
Check and Diagnose: use right-triangle trigonometric ratios

ERROR ANALYSIS

- Opposite and adjacent depend on the chosen reference angle.
- Keep the calculator in the requested angle mode.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?



Change the angle and opposite and adjacent trade places.

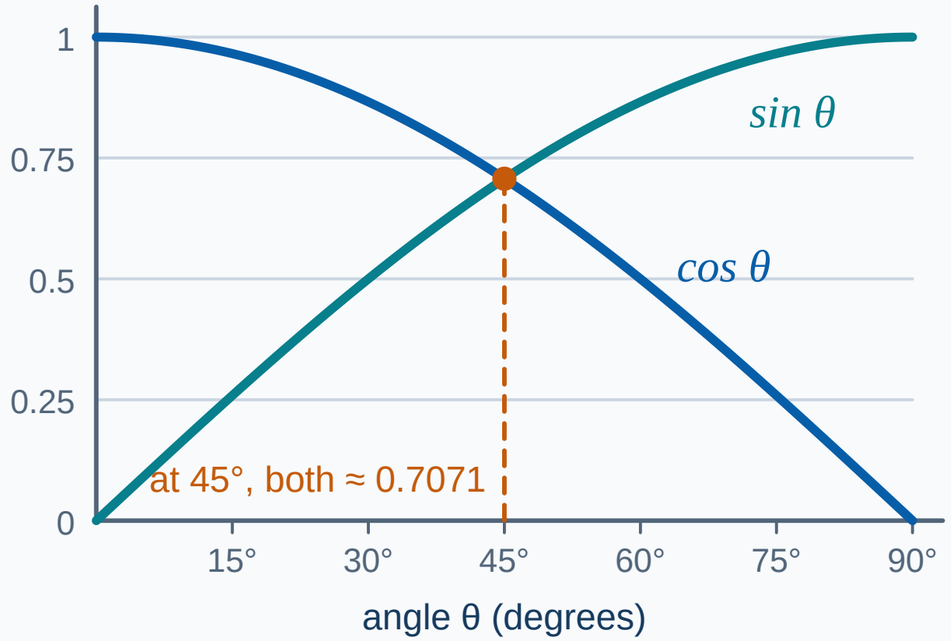
Source: 16, 17

Make It Stick: use right-triangle trigonometric ratios

PACING CHECKPOINT

Write the ratio-selection rule you will use under time pressure.

Exit response
Answer in complete mathematical sentences and identify one remaining question.



For acute angles, sine and cosine stay between 0 and 1.

Predict: choose the ratio for a word problem

PREDICT FIRST

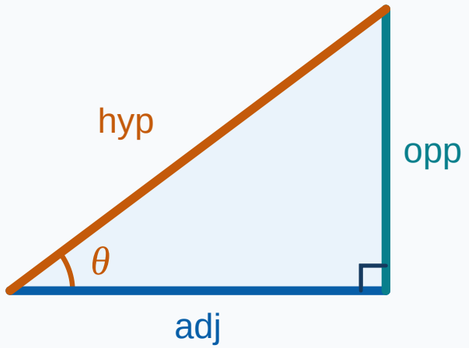
You know the hypotenuse and want the side opposite the angle. Which ratio do you use, and do you multiply or divide?

Commit to a response before discussion.

Choose the ratio from the side you know and the side you want

CONCEPT

- Use the ratio that contains the side you know and the side you want.
- If the unknown is on top of the fraction, multiply; if it is on the bottom, divide.
- To find an angle, use inverse trig: $A = \sin^{-1}\left(\frac{\text{opp}}{\text{hyp}}\right)$.
- To solve a right triangle, find all three missing parts, then check the angle sum and the Pythagorean theorem.



know and want	use
hyp & opp	**sin**
hyp & adj	**cos**
adj & opp	**tan**

Unknown on top: multiply.

Unknown on the bottom: divide.

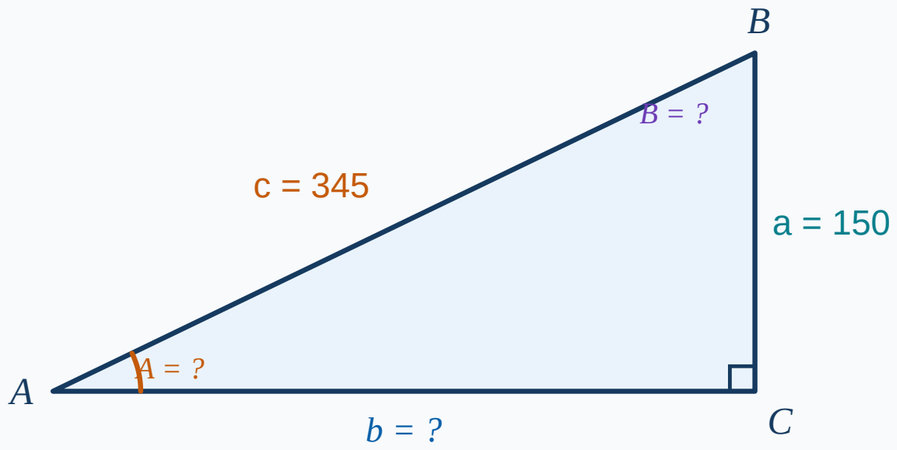
Pick the ratio from the two sides in the problem.

Source: 19, 20

Model the Skill: solve a right triangle

Solve right triangle ABC with $C = 90^\circ$, $a = 150$ and $c = 345$: find b, A and B.

1. $\sin A = \frac{150}{345}$, so $A = \sin^{-1}\left(\frac{150}{345}\right) \approx 25.8^\circ$. Type the parentheses: $\sin^{-1}(150 \div 345)$.
2. $B = 90^\circ - A \approx 64.2^\circ$, and $b = \sqrt{345^2 - 150^2} \approx 310.7$.
3. Check: $25.8^\circ + 64.2^\circ + 90^\circ = 180^\circ$ and $150^2 + 310.7^2 \approx 345^2$.

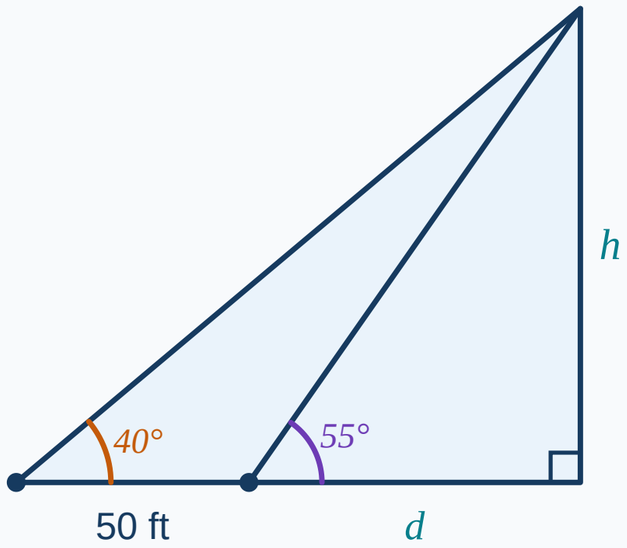


Drawn to scale. Find the side b and both acute angles.

Model the Skill: two right triangles, one height

From one point the angle of elevation to the top of a tower is 40°. From a point 50 ft closer it is 55°. Find the height h.

1. Near triangle: $h = d\tan 55^\circ$. Far triangle: $h = (d + 50)\tan 40^\circ$.
2. Set them equal and solve:
$$d = \frac{50\tan 40^\circ}{\tan 55^\circ - \tan 40^\circ} \approx 71.23 \text{ ft.}$$
3. $h = d\tan 55^\circ \approx 101.72$ ft. Check: $(71.23 + 50)\tan 40^\circ \approx 101.72$ ft.



Drawn to scale. Both triangles share the height h.

Source: Authored

You Try: an unknown in the denominator

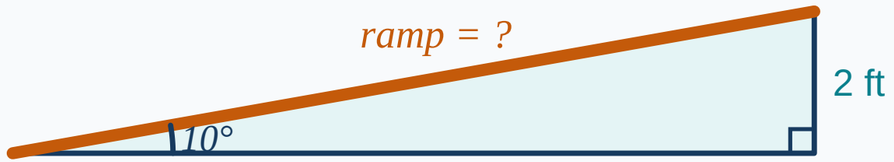
STUDENT PRACTICE

A ramp must rise 2 ft, and it makes a 10° angle with the ground. How long is the ramp?

Reveal solution

FEEDBACK AND SOLUTION

$\sin 10^\circ = 2/L$, so $L = 2 \div \sin 10^\circ \approx 11.52$ ft.



Drawn to scale: a 10° ramp is long and low.

The ramp is the hypotenuse; 2 ft is opposite the 10° angle.

Source: 22, 23

Check and Diagnose: right-triangle word problems

ERROR ANALYSIS

- A leg must be shorter than the hypotenuse.
- Degree mode, and parentheses for inverse trig: $\sin^{-1}(5.5 \div 6.5)$, not $\sin^{-1} 5.5 \div 6.5$.
- In two-triangle problems, keep d and $d + 50$ with the right angles, and round only at the end.

Self-check

What evidence confirms the setup, sign, unit, or magnitude?

Make It Stick: right-triangle word problems

PACING CHECKPOINT

List the steps to solve a right triangle and the two checks you finish with.

Exit response
Answer in complete mathematical sentences and identify one remaining question.

Session Summary and Exit Ticket

1. State the most important idea from today without looking at your notes.
2. Complete one representative setup and identify the step most likely to cause an error.
3. Write one question that should be answered before the next assessment.

Exit response

Answer in complete mathematical sentences and identify one remaining question.

Reveal review guidance

FEEDBACK AND SOLUTION

Use the objective roadmap to identify any skill that still needs deliberate practice.